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FOUNDATIONS OF ALGEBRAIC GEOMETRY AND COMMUTATIVE ALGEBRA

A SUMMER INTERNSHIP REPORT

Submitted in partial fulfillment of the academic requirements for the
Summer Internship Program

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JULY, 2026

DEPARTMENT OF MATHEMATICAL SCIENCES
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CANDIDATE'S DECLARATION

I, **RATUL DUTTA**, Roll No. **492154**, student of **M.Sc. Mathematics and Computing**, DST-CIMS, Institute of Science, Banaras Hindu University, hereby declare that the summer internship report entitled:

"FOUNDATIONS OF ALGEBRAIC GEOMETRY AND COMMUTATIVE ALGEBRA"

submitted by me to **Dr. Anoop Singh**, Department of Mathematical Sciences, Indian Institute of Technology (BHU), Varanasi, is a record of original training and study carried out by me. This work has not previously formed the basis for the award of any Degree, Diploma, Associateship, Fellowship or other similar title or recognition.

Place: Varanasi **Student Signature**

Date: 24 July 2026 **RATUL DUTTA**

DEPARTMENT OF MATHEMATICAL SCIENCES
INDIAN INSTITUTE OF TECHNOLOGY (BHU)
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CERTIFICATE

I hereby certify that the Summer Internship Report entitled

“FOUNDATIONS OF ALGEBRAIC GEOMETRY AND COMMUTATIVE ALGEBRA”

submitted by **RATUL DUTTA**, Roll No. **492154**, student of **M.Sc. Mathematics and Computing**, DST-CIMS, Institute of Science, Banaras Hindu University, is a record of the internship work carried out by the student under my supervision during the period of May 18, 2026 to July 18, 2026.

Place: Varanasi

Date: 24 July 2026

(Dr. Anoop Singh)
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ABSTRACT

This report details the work and independent mathematical study completed during an eight-week summer internship focusing on the foundational structures of Commutative Algebra and their profound geometric interpretations in Algebraic Geometry. The theoretical progression builds iteratively, beginning with algebraic preliminaries—including rings, modules, and K-algebras—towards the robust structural frameworks of Noetherian and Artinian chain conditions. Key analytical milestones encompass the consequences of Hilbert's Basis Theorem, the algebraic formalization of localization, and the critical insights into local module properties provided by the Krull-Nakayama Lemma. The climax of the study establishes the rigid, bidirectional dictionary between pure algebra and affine geometry via Hilbert's Nullstellensatz. This translates geometrically into the formalization of the Zariski topology and algebraic sets, ultimately concluding with the dimensional analysis granted by integral extensions and the Noether Normalization Lemma.

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Student Signature

RATUL DUTTA

Contents

LIST OF SYMBOLS

Symbol	Meaning
Rings, Fields, and Ideals	
K	A field
A,R	A commutative ring with unity
I,J	Ideals in a ring
$\mathfrak{I}(A)$	The set of all ideals of a ring A
$r\text{-}\mathfrak{I}(A)$	The set of all radical ideals of a ring A
$\mathfrak{p}$	A prime ideal
$\mathfrak{m}$	A maximal ideal
$\mathfrak{n}_A, \mathrm{nil}(A)$	The nilradical (or set of all nilpotent elements) of a ring A
$\mathfrak{m}_A, J(A)$	The Jacobson radical of a ring A
A/I	Quotient ring (or residue class ring) of A modulo I
$\sqrt{I}$	The radical of an ideal I
Algebraic Geometry and Spectra	
$A[x_1,\ldots,x_n]$	Polynomial ring in n variables over A
$\mathbb{A}^n_K$	Affine n-space over the field K
$V_L(I)$	The vanishing locus (or zero set) of an ideal I over L
$I_K(X)$	The ideal of polynomials vanishing on an algebraic set X
$\mathrm{Spec}(A)$	The spectrum (set of all prime ideals) of a ring A
$\mathrm{Spm}(A)$	The maximal spectrum (set of all maximal ideals) of a ring A
$K\text{-}\mathrm{Spec}(A)$	The K-spectrum (set of K-rational points) of a K-algebra A
$D(\mathfrak{a}), D(f)$	Distinguished (or principal) open sets in the Zariski topology

Modules and Localization

M, N, V	Modules over a ring
$\mathscr{S}_A(V)$	The set of all A-submodules of an A-module V
$\ker(f)$	Kernel of a homomorphism f
$\mathrm{im}(f)$	Image of a homomorphism f
$M \oplus N$	Direct sum of modules M and N
$S^{-1}A$	Ring of fractions of A with respect to a multiplicatively closed set S
$S^{-1}M$	Localization of an A-module M with respect to S

Algebras and Homomorphisms

$Z(B)$	The center of an algebra B
A-alg	The collection (or category) of all A-algebras
$\mathrm{Hom}_{A\text{-alg}}(B, C)$	The set of all A-algebra homomorphisms from B to C
ε_x	The substitution (or evaluation) homomorphism

CHAPTER 1
INTRODUCTION

1.1 Overview

The study of Commutative Algebra serves as the fundamental machinery for Algebraic Geometry. Historically, geometric intuition provided the motivation for many abstract algebraic concepts, while the rigorous framework of algebra allowed for the generalization and proof of profound geometric statements. During this eight-week summer internship, the focus was to deeply understand this beautiful dictionary between algebraic structures (specifically commutative rings and ideals) and geometric spaces (affine varieties). By translating geometric problems into purely algebraic ones, one can leverage powerful algebraic tools—like localization, radicals, and chain conditions—to unravel the topological properties of spaces defined by polynomial equations.

The exposition in this report largely follows the standard treatments of [1] and [2], supplemented by lecture notes from [3].

1.2 Objectives

The primary objectives of this structured internship study were:

1. To establish a rigorous mathematical foundation in commutative rings, ideals, and modules, acting as the bedrock for the study of advanced K-algebras.

2. To understand the algebraic properties of Noetherian and Artinian structures, specifically focusing on how ascending and descending chain conditions restrict the size and complexity of modules and rings.

3. To analyze the structural importance of localization, radical ideals, and local-global theorems such as the Krull-Nakayama Lemma.

4. To explore the geometric correspondence of algebraic sets via Hilbert's Nullstellensatz, fully realizing the bidirectional bridge between radical ideals and affine spaces.

5. To formalize the Zariski topology on arbitrary spectra and affine spaces, concluding with dimensional analysis through integral extensions and the Noether Normalization Lemma.

CHAPTER 2

ALGEBRAIC PRELIMINARIES

2.1 Rings and Ideals

Definition 2.1.1 (Ring, Integral Domain, and Field)**.** Throughout this report, the term *ring* shall denote a commutative ring with a multiplicative identity $1 \neq 0$. An *integral domain* is a non-zero ring with no zero divisors (i.e., if $ab = 0$, then $a = 0$ or $b = 0$). A *field* is a ring in which every non-zero element has a multiplicative inverse.

Definition 2.1.2 (Ideal)**.** An *ideal* I of a ring A is an additive subgroup of A such that for any $a \in A$ and $x \in I$, the product $ax \in I$. Ideals are the central objects of study in commutative algebra because they correspond to the kernels of ring homomorphisms.

Example 2.1.3 (Ideal)**.** Consider the ring of integers $A = \mathbb{Z}$. Let $I = 2\mathbb{Z}$ be the set of all even integers. First, $2\mathbb{Z}$ forms a subring as it is closed under addition, subtraction, and multiplication. Furthermore, for any arbitrary integer $a \in \mathbb{Z}$ and any even integer $x = 2k \in 2\mathbb{Z}$ (where $k \in \mathbb{Z}$), their product is $ax = a(2k) = 2(ak)$. Since ak is an integer, the product $ax \in 2\mathbb{Z}$. Thus, $2\mathbb{Z}$ "absorbs" multiplication from any element in the parent ring $\mathbb{Z}$, making it a valid ideal.

Definition 2.1.4 (Residue Class Ring)**.** Let I be an ideal of a ring A. The *residue class ring* (or quotient ring) A/I inherits a natural ring structure, effectively "killing" the elements of the ideal.

Example 2.1.5 (Residue Class Ring)**.** Building on the previous example, let $A = \mathbb{Z}$ and the ideal $I = 2\mathbb{Z}$. The residue class ring (or quotient ring) $\mathbb{Z}/2\mathbb{Z}$ partitions the integers into disjoint equivalence classes (cosets) of the form $a + 2\mathbb{Z}$. There are exactly two such distinct classes:

- $\overline{0} = 0 + 2\mathbb{Z}$ (the set of all even integers)
- $\overline{1} = 1 + 2\mathbb{Z}$ (the set of all odd integers)

This quotient ring $\mathbb{Z}/2\mathbb{Z} = \{\overline{0}, \overline{1}\}$ inherits its ring structure from $\mathbb{Z}$, where operations are defined on the representatives modulo 2 (e.g., $\overline{1} + \overline{1} = \overline{0}$). By "killing" the elements of the ideal (making every even number equivalent to zero), we are left with a fundamental and structurally simpler ring.

Definition 2.1.6 (Ring Homomorphism)**.** A *ring homomorphism* is a map $\phi : A \to B$ between two rings that preserves the fundamental operations: $\phi(x+y) = \phi(x) + \phi(y)$, $\phi(xy) = \phi(x)\phi(y)$, and $\phi(1_A) = 1_B$.

Theorem 2.1.7 (Fundamental Theorem of Homomorphism)**.** *Let $\phi : A \to B$ be a ring homomorphism and I be an ideal in A with $I \subseteq \ker\phi$. Then there exists a unique ring homomorphism $\overline{\phi} : A/I \to B$ such that $\overline{\phi} \circ \pi = \phi$, where $\pi : A \to A/I$ is the canonical projection map.*

This means the following diagram is commutative:

$$\begin{tikzcd} A \arrow[r, "\phi"] \arrow[d, "\pi"'] & B \\ A/I \arrow[ur, dashed, "\overline{\phi}"'] & \end{tikzcd}$$

Example 2.1.8 (Canonical Projection)**.** Consider the ring of integers $\mathbb{Z}$ and the residue class ring $\mathbb{Z}/n\mathbb{Z}$ for some integer $n > 1$. The map $\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ defined by mapping each integer to its equivalence class modulo n:

$$\pi(a) = \overline{a} = a + n\mathbb{Z}$$

is a surjective ring homomorphism. It preserves both fundamental operations, as $\pi(a+b) = \overline{a+b} = \overline{a} + \overline{b} = \pi(a) + \pi(b)$ and $\pi(ab) = \overline{ab} = \overline{a}\overline{b} = \pi(a)\pi(b)$. Furthermore, it maps the multiplicative identity $1 \in \mathbb{Z}$ to the identity $\overline{1} \in \mathbb{Z}/n\mathbb{Z}$. The kernel of this homomorphism is the ideal $n\mathbb{Z}$.

Example 2.1.9 (Evaluation Homomorphism)**.** Let R be a commutative ring and $R[x]$ be the polynomial ring in one variable over R. Choose a fixed element $c \in R$. The evaluation map $\varepsilon_c : R[x] \to R$ defined by substituting c into the polynomial:

$$\varepsilon_c(F(x)) = F(c)$$

is a ring homomorphism. For instance, if $F(x) = x^2 + 3x + 2$, then $\varepsilon_c(F) = c^2 + 3c + 2$. This map naturally preserves the ring operations because evaluating a sum or product of polynomials at c yields the exact same result as the sum or product of their individual evaluations.

2.2 Prime and Maximal Ideals

Definition 2.2.1. An ideal $\mathfrak{p}$ in a ring A is called a *prime ideal* if $\mathfrak{p} \neq A$ and whenever $xy \in \mathfrak{p}$ for some $x, y \in A$, then either $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. Equivalently, $A/\mathfrak{p}$ is an integral domain.

Example 2.2.2. In the ring of integers $\mathbb{Z}$, the ideal $(p) = p\mathbb{Z}$ generated by a prime number p is a prime ideal. If $ab \in (p)$, then p divides ab, implying p divides a or p divides b, so $a \in (p)$ or $b \in (p)$.

Example 2.2.3 (Non-example)**.** In $\mathbb{Z}$, the ideal $(6) = 6\mathbb{Z}$ is not prime. We observe that $2 \cdot 3 = 6 \in (6)$, but neither $2 \in (6)$ nor $3 \in (6)$.

Definition 2.2.4. An ideal $\mathfrak{m}$ is called a *maximal ideal* if $\mathfrak{m} \neq A$ and there is no proper ideal I strictly containing $\mathfrak{m}$. Equivalently, $A/\mathfrak{m}$ is a field. Every maximal ideal is automatically a prime ideal since every field is an integral domain.

Example 2.2.5. In the polynomial ring $\mathbb{R}[x]$, the ideal (x) is maximal. The quotient ring $\mathbb{R}[x]/(x)$ is isomorphic to the field of real numbers $\mathbb{R}$.

Example 2.2.6 (Non-example)**.** Consider the polynomial ring in two variables $\mathbb{R}[x,y]$. The ideal generated by x, denoted (x), is prime because the quotient $\mathbb{R}[x,y]/(x) \cong \mathbb{R}[y]$ is an integral domain. However, (x) is not maximal because it is strictly contained in the proper ideal (x,y), which represents polynomials with no constant term.

2.3 Krull's Theorem

To guarantee the existence of maximal ideals in arbitrary rings, one relies on the Axiom of Choice, often applied in the form of Zorn's Lemma.

Theorem 2.3.1 (Krull's Theorem)**.** *Let A be a ring and $I \neq A$ be a proper ideal. Then there exists a maximal ideal $\mathfrak{m}$ of A such that $I \subseteq \mathfrak{m}$.*

Corollary 2.3.2. *Every non-unit element in a ring is contained in some maximal ideal.*

This simple fact is incredibly powerful, particularly when exploring the spectrum of a ring.

2.4 Affine Algebraic Sets

Definition 2.4.1. Let K be a field. Affine n-space over K, denoted $\mathbb{A}^n_K$, is the set of all n-tuples $(a_1,\ldots,a_n)$ with $a_i \in K$. The vanishing locus associated with a subset $S \subseteq K[X_1,\ldots,X_n]$ is defined as:

$$V_K(S) = \{(a_1,\ldots,a_n) \in \mathbb{A}^n_K \mid f(a_1,\ldots,a_n) = 0 \text{ for all } f \in S\}$$

A subset $X \subseteq \mathbb{A}^n_K$ is called an affine algebraic set if $X = V_K(I)$ for some ideal $I \subseteq K[X_1,\ldots,X_n]$.

2.5 Modules and Submodules

A module over a ring is a generalization of a vector space over a field.

Definition 2.5.1. Let A be a ring. An A-module is an abelian group $(V,+)$ equipped with a scalar multiplication operation $A \times V \to V$ that satisfies distributive and associative axioms identical to those of vector spaces.

Definition 2.5.2 (A-module Homomorphism)**.** Let V and W be two A-modules. A map $f : V \to W$ is called an A-module homomorphism (or A-linear map) if:

1. $f(v+v') = f(v) + f(v')$ for all $v, v' \in V$
2. $f(av) = af(v)$ for all $a \in A$ and $v \in V$

The set of all A-module homomorphisms from V into W is denoted by $\mathrm{Hom}_A(V,W)$.

Definition 2.5.3 (Submodules)**.** A subset $V' \subseteq V$ is called an A-submodule of V if $(V',+)$ is a subgroup of $(V,+)$ and it is closed under scalar multiplication, i.e., $a \cdot v' \in V'$ for all $a \in A$ and $v' \in V'$.

Some key properties and examples of submodules include:

1. Ideals in a ring A are A-submodules of the A-module A.

2. Given any element $v \in V$, the set $Av := \{av \mid a \in A\} \subseteq V$ is an A-submodule of V. This is the smallest A-submodule of V containing v.

3. Let $v_1, v_2, \ldots, v_n \in V$. The smallest A-submodule of V containing all $v_1, \ldots, v_n$ is:

$$Av_1 + \cdots + Av_n := \{a_1 v_1 + \cdots + a_n v_n \mid a_1, \ldots, a_n \in A\}$$

   This submodule is said to be generated by $v_1, v_2, \ldots, v_n$.

4. More generally, for an arbitrary family v_i where $i \in B$ in V, the smallest submodule containing all elements in this arbitrary family is:

$$\sum_{i \in B} Av_i := \left\{ \sum_{j \in J} a_j v_j \,\middle|\, J \subseteq B \text{ is finite}, a_j \in A \right\}$$

Example 2.5.4. Let $A = \mathbb{Z}$, and consider the elements $4, 6 \in \mathbb{Z}$. The $\mathbb{Z}$-submodule of $\mathbb{Z}$ generated by 4 and 6 is $\mathbb{Z}4 + \mathbb{Z}6 = \mathbb{Z}2$. Note that the generating set is not unique, as the same submodule can be generated simply by 2.

CHAPTER 3

BRIDGE TO GEOMETRY: POLYNOMIAL ALGEBRAS, RATIONAL POINTS, AND SPECTRA

3.1 Algebras over Fields and Rings

Definition 3.1.1 (K-Algebra)**.** Let K be a base field. An *algebra over K* (or K-algebra) can be defined in two equivalent ways:

1. A commutative ring $(B,+,\cdot)$ that is simultaneously a K-vector space $(B,+)$, such that the ring multiplication and vector space scalar multiplication are compatible:

$$(ax)\cdot(by) = (ab)(x\cdot y) \quad \forall a,b \in K \text{ and } \forall x,y \in B$$

2. A pair (B,ϕ), where B is a commutative ring and $\phi : K \to B$ is a ring homomorphism defined by $a \mapsto a\cdot 1_B$. This homomorphism is called the *structure homomorphism* of the K-algebra B.

This concept generalizes naturally from base fields to arbitrary commutative rings.

Definition 3.1.2 (A-Algebra)**.** Let A be a commutative ring. An *A-algebra* is a pair (B,ϕ), where B is a ring (not necessarily commutative) and $\phi : A \to B$ is a ring homomorphism (the structure homomorphism), subject to the extra condition that the image of A lies in the center of B:

$$\phi(A) \subseteq Z(B) = \{b \in B \mid bc = cb \text{ for all } c \in B\}$$

If the ring B is commutative, this center condition is trivially satisfied.

Example 3.1.3.

1. The polynomial ring $A[x_1,\dots,x_n]$ is an A-algebra via the natural inclusion homomorphism $i : A \to A[x_1,\dots,x_n]$ defined by $a \mapsto a$. The algebra pair is formally denoted $(A[x_1,\dots,x_n],i)$.

2. Let $\mathfrak{a} \subseteq A[x_1,\dots,x_n]$ be an ideal, and let $\pi : A[x_1,\dots,x_n] \to A[x_1,\dots,x_n]/\mathfrak{a}$ be the canonical projection mapping $F \mapsto F+\mathfrak{a} = \overline{F}$. Then the quotient ring is an A-algebra equipped with the

composed structure homomorphism $\pi \circ i$. We denote this A-algebra as $(A[x_1,\dots,x_n]/\mathfrak{a}, \pi \circ i)$.

$$\begin{tikzcd}
A \arrow[r, "i"] \arrow[dr, "\pi\circ i"'] & A[x_1,\dots,x_n] \arrow[d, "\pi"] \\
& A[x_1,\dots,x_n]/\mathfrak{a}
\end{tikzcd}$$

Definition 3.1.4 (A-Algebra Homomorphism)**.** Let (B,ϕ) and (C,ψ) be two A-algebras. A map $f : B \to C$ is called an *A-algebra homomorphism* if it satisfies two conditions:

1. f is a ring homomorphism (specifically, $1_B \mapsto 1_C$).
2. f is an A-module homomorphism (i.e., it is additive and A-linear with respect to scalar multiplication).

Equivalently, f is an A-algebra homomorphism if the following diagram is commutative, meaning $\psi = f \circ \phi$:

$$\begin{tikzcd}
& A \arrow[dl, "\phi"'] \arrow[dr, "\psi"] & \\
B \arrow[rr, "f"'] & & C
\end{tikzcd}$$

We introduce the following notations and properties regarding the category of A-algebras:

- A-alg denotes the collection of all A-algebras.
- $\mathrm{Hom}_{A\text{-alg}}(B,C)$ denotes the set of all A-algebra homomorphisms from A-algebra B to A-algebra C.
- The composition of two A-algebra homomorphisms is also an A-algebra homomorphism (e.g., if $B \xrightarrow{f} C \xrightarrow{g} D$, then $g \circ f$ is an A-algebra homomorphism).
- The identity map on any A-algebra is an A-algebra homomorphism.

3.2 Universal Property of Polynomial Algebras

Let A be a ring and B be an A-algebra. Given elements $b_1,\dots,b_n \in B$, let $b = (b_1,\dots,b_n)$. Then there exists a unique A-algebra homomorphism:

$$\varepsilon_b : A[x_1,\dots,x_n] \to B$$

such that $x_i \mapsto b_i$ for $i = 1,\dots,n$. Consequently, for any polynomial F, the map evaluates as:

$$F(x_1,\dots,x_n) \mapsto F(b_1,\dots,b_n)$$

This unique homomorphism ε_b is called the *substitution homomorphism* or *evaluation map*.

Example 3.2.1. Let $A = \mathbb{Z}$ and $n = 1$, giving the polynomial ring $\mathbb{Z}[x]$. Let the target A-algebra be $B = \mathbb{Q}$. An evaluation map $\varepsilon_x : \mathbb{Z}[x] \to \mathbb{Q}$ acts on polynomials by substituting the variable, such that $x \mapsto x$, $2x^2 \mapsto 2x^2$, and $2x^3 + 2x \mapsto 2x^3 + 2x$.

In the specific case where the base ring is a field K, the polynomial ring $K[X_1,\dots,X_n]$ is a K-algebra of finite type (where $X_1,\dots,X_n$ is a set of K-algebra generators). However, it is important to note that it is not a finite-dimensional K-vector space.

3.3 The K-Spectrum of a K-Algebra

For a polynomial ring $K[X_1,\dots,X_n]$ over a base field K, there is a natural correspondence between points in the affine space K^n and certain maximal ideals. Specifically, we have the following sequence of mappings:

$$K^n \longrightarrow \mathrm{Hom}_{K\text{-alg}}(K[X_1,\dots,X_n],K) \longrightarrow K\text{-Spec}(K[X_1,\dots,X_n])$$

A point $a=(a_1,\dots,a_n)\in K^n$ corresponds to the substitution (or evaluation) homomorphism:

$$\varepsilon_a : K[X_1,\dots,X_n] \xrightarrow{K\text{-alg homo}} K$$

defined by $X_i \mapsto a_i$ and $F(X_1,\dots,X_n)\mapsto F(a_1,\dots,a_n)$.

The kernel of this homomorphism is a maximal ideal $\mathfrak{m}_a$:

$$\mathrm{Ker}(\varepsilon_a) = \langle X_1-a_1,\dots,X_n-a_n\rangle = \mathfrak{m}_a$$

This ideal $\mathfrak{m}_a$ is an element of $\mathrm{Spm}(K[X_1,\dots,X_n]) \subset \mathrm{Spec}(K[X_1,\dots,X_n])$.

More generally, let A be a K-algebra of finite type. This means there exists a surjective K-algebra homomorphism:

$$\varepsilon_x : K[X_1,\dots,X_n] \twoheadrightarrow A = K[x_1,\dots,x_n]$$

mapping $X_i \mapsto x_i$, where $x_1,\dots,x_n$ are the K-algebra generators for A. By the First Isomorphism Theorem, we have an isomorphism of K-algebras:

$$K[X_1,\dots,X_n]/\mathrm{Ker}(\varepsilon_x) \cong A$$

where $\mathrm{Ker}(\varepsilon_x)=\mathfrak{a}$ represents the ideal of polynomial relations over K among the generators $x_1,\dots,x_n$. (Note that $\mathfrak{a}$ is generally not maximal, as A is not necessarily a field).

We formally define the K-spectrum of A as:

$$K\text{-Spec}(A) = \{\mathfrak{m}\in \mathrm{Spm}(A) \mid A/\mathfrak{m}\cong K\}$$

where $A/\mathfrak{m}$ represents the residue field.

3.4 K-Rational Points

The concept of K-rational points provides a formal bridge between the algebraic object K-$\mathrm{Spec}(A)$ and the geometric points in affine space.

Definition 3.4.1 (K-Rational Point)**.** A point is considered a *K-rational point* if its corresponding maximal ideal is generated entirely by linear polynomials:

$$\mathfrak{m}_a = \langle X_1 - a_1, \ldots, X_n - a_n \rangle$$

where $a_1, \ldots, a_n \in K$.

For a K-algebra of finite type, represented as a quotient $A = K[X_1, \ldots, X_n]/I$, evaluating a K-rational point is equivalent to finding a tuple $a = (a_1, \ldots, a_n) \in K^n$ that evaluates to zero for every polynomial in the ideal I.

Geometrically, this establishes a natural bijection between the algebraic K-rational points of $K[X_1, \ldots, X_n]/I$ and the geometric points of the affine algebraic set $V(I) \subseteq \mathbb{A}^n_K$:

$$K\text{-Spec}(K[X_1, \ldots, X_n]/I) \longleftrightarrow V(I)$$

Through this correspondence, the study of geometric solutions to polynomial equations becomes equivalent to studying the maximal ideals of the algebra whose residue field is identically K.

3.5 Radical Ideals and the Vanishing Locus

Let A be an arbitrary commutative ring.

Definition 3.5.1 (Radical Ideal)**.** An ideal $\mathfrak{a}$ in A (where $\mathfrak{a} \in \mathfrak{I}(A)$) is called a radical ideal if $\mathfrak{a} = \sqrt{\mathfrak{a}}$.

The expression $\sqrt{\mathfrak{a}}$ denotes the radical of the ideal, defined as $\{f \in A \mid f^r \in \mathfrak{a} \text{ for some } r \in \mathbb{N}\}$. This radical is itself an ideal in A, and it is always true that $\mathfrak{a} \subseteq \sqrt{\mathfrak{a}}$. Consequently, $\mathfrak{a} = \sqrt{\mathfrak{a}}$ if and only if for all $f \in A$, if $f^r \in \mathfrak{a}$ then $f \in \mathfrak{a}$ for some $r > 0$.

A crucial property in algebraic geometry is that the vanishing locus $V_L(I)$ depends only on the "radical of the ideal I" and not on I itself.

Example 3.5.2. Consider the fields $K = \mathbb{C} = L$, dimension $n = 2$, and the space $L^2 = \mathbb{C}^2$ corresponding to the polynomial ring $\mathbb{C}[Z, W]$.

- Let $I_1 = \langle Z \rangle$ and $I_2 = \langle Z^2 \rangle$.
- While the ideals are distinct ($I_1 \neq I_2$), their vanishing loci are identical: $V_{\mathbb{C}}(I_1) = V_{\mathbb{C}}(I_2) = V_{\mathbb{C}}(I)$.
- Geometrically, both loci represent the W-axis, although $V_{\mathbb{C}}(Z^2)$ is conceptually "thicker for Z^2".
- This principle extends generally to $I = \langle Z^m \rangle = I_1^m$.

3.6 Field Extensions and Geometric Contexts

Let K be a field, and L/K be a field extension with L algebraically closed. The choice of these fields defines different domains of mathematical study:

- When $K = \mathbb{R}$ and $L = \mathbb{C}$, this forms the basis of classical algebraic geometry.
- When $K = \mathbb{Q}$ and $L = \mathbb{C}$ or $L = \overline{\mathbb{Q}}$ (the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$), this represents arithmetic geometry.
- The field $\overline{\mathbb{Q}}$ is the field of algebraic numbers, defined as $\{z \in \mathbb{C} \mid f(z) = 0$ for some monic $0 \neq f \in \mathbb{Q}[x]\}$.
- Arithmetic geometry, which studies these questions over number fields and their rings of integers, remains an active and rapidly developing area of research.
- When $K = \mathbb{F}$ is a finite field and $L = \overline{\mathbb{F}}$, this setting underlies applications in coding theory and cryptography, where polynomial equations over finite fields play a central role.

CHAPTER 4

NOETHERIAN STRUCTURES AND HILBERT'S BASIS THEOREM

4.1 Noetherian and Artinian Ordered Sets

Let $(X,\leq)$ be an ordered set.

Definition 4.1.1 (Noetherian Ordered Set)**.** An ordered set X is called Noetherian if every non-empty subset $Y \subseteq X$ has a maximal element. An element $y_0 \in Y$ is called a maximal element for Y if there is no $y \in Y$ such that $y > y_0$.

Note that a maximal element is not necessarily unique, and a Noetherian ordered set need not be totally ordered.

Example 4.1.2. 1. Finite ordered sets are Noetherian.

2. The set of natural numbers $\mathbb{N}$ with the natural order is not Noetherian.

Definition 4.1.3 (Sequences in Ordered Sets)**.** Let $(X,\leq)$ be an arbitrary ordered set.

- A sequence $(x_n)_{n\in\mathbb{N}}$ in X is called stationary if there exists $n_0 \in \mathbb{N}$ such that $x_n = x_{n_0}$ for all $n \geq n_0$.
- A sequence $(x_n)_{n\in\mathbb{N}}$ in X is called ascending (or descending) if for all $i, j \in \mathbb{N}$ with $i \leq j$, we have $x_i \leq x_j$ (or $x_i \geq x_j$).

Definition 4.1.4 (Artinian Ordered Set)**.** An ordered set $(X,\leq)$ is called Artinian if every non-empty subset $Y \subseteq X$ has a minimal element. That is, there exists $y_0 \in Y$ such that for every $y \in Y$, if $y \leq y_0$, then $y = y_0$.

Corollary 4.1.5. *An ordered set $(X,\leq)$ is Artinian if and only if every descending sequence on X is stationary. This is equivalent to saying that X satisfies the Descending Chain Condition (DCC).*

Proposition 4.1.6. *Let $(X,\leq)$ be an ordered set. The following properties hold:*

1. *Any finite ordered set is Artinian.*
2. *$(X,\leq)$ is Noetherian (or Artinian) if and only if the opposite ordered set $(X,\leq^{op})$ is Artinian (or Noetherian).*

4.2 Noetherian Modules and Rings

Let A be a ring and let V be an A-module. Consider $\mathscr{S}_A(V)$, the set of all A-submodules of V ordered by the natural inclusion $\subseteq$. Thus, $(\mathscr{S}_A(V), \subseteq)$ forms an ordered set.

Note that if we consider the ring A as an A-module over itself, $\mathscr{S}_A(A)$ is precisely the set of all ideals of A, often denoted $\mathfrak{I}(A)$.

Definition 4.2.1 (Noetherian and Artinian Modules)**.** An A-module V is called Noetherian (respectively, Artinian) if the ordered set $(\mathscr{S}_A(V), \subseteq)$ is Noetherian (respectively, Artinian).

Equivalently, V is Noetherian (Artinian) if and only if:

- Every non-empty family of submodules of V has a maximal (minimal) element.
- Every ascending (descending) chain in $\mathscr{S}_A(V)$ is stationary. In this case, we say V satisfies the Ascending Chain Condition, ACC (Descending Chain Condition, DCC).

Lemma 4.2.2. *Let A be a ring and V be an A-module. Then V is a Noetherian A-module if and only if every A-submodule of V is finitely generated.*

Definition 4.2.3 (Noetherian and Artinian Rings)**.** Let A be a ring. Then A is called a Noetherian (respectively, Artinian) ring if the A-module A is Noetherian (respectively, Artinian).

Equivalently:

- A is a Noetherian ring $\iff$ A has the ACC on ideals (i.e., any ascending chain of ideals $\mathfrak{a}_0 \subseteq \mathfrak{a}_1 \subseteq \cdots \subseteq \mathfrak{a}_m \subseteq \cdots \subseteq A$ is stationary) $\iff$ $(\mathfrak{I}(A), \subseteq)$ is a Noetherian ordered set.
- A is an Artinian ring $\iff$ A has the DCC on ideals $\iff$ $(\mathfrak{I}(A), \subseteq)$ is an Artinian ordered set.

In particular, a ring A is Noetherian if and only if every ideal in A is finitely generated.

A primary structural goal in commutative algebra is to establish that the polynomial ring $K[x_1, \ldots, x_n]$ over a field K is Noetherian.

4.3 Properties of Noetherian and Artinian Modules

Let A be a ring and V be an A-module, leading to the ordered set $(\mathscr{S}_A(V), \subseteq)$. For any submodule $U \subseteq V$, we denote the residue class module (or quotient module) of V by U as V/U.

Lemma 4.3.1. *Let $U \subseteq V$ be an A-submodule of V. Then, V is Noetherian (respectively, Artinian) if and only if both U and V/U are Noetherian (respectively, Artinian).*

This fundamental lemma yields several important corollaries.

Corollary 4.3.2. *Let $f : V \to W$ be an A-module homomorphism. Then V is Noetherian (respectively, Artinian) if and only if both $\ker f$ and $Im f$ are Noetherian (respectively, Artinian).*

Corollary 4.3.3. *Let $0 \to V' \xrightarrow{f'} V \xrightarrow{f} V'' \to 0$ be a short exact sequence of A-modules and A-module homomorphisms. This sequence being exact implies three conditions: f' is injective, $Im f' = \ker f$, and f is surjective. Then V is Noetherian (respectively, Artinian) if and only if both V' and V'' are Noetherian (respectively, Artinian).*

Corollary 4.3.4. *A finite direct sum of Noetherian (respectively, Artinian) A-modules is again a Noetherian (respectively, Artinian) A-module.*

Corollary 4.3.5. *Let V be an A-module and let $V_1,\dots,V_n \subseteq V$ be A-submodules of V. If all $V_1,\dots,V_n$ are Noetherian (respectively, Artinian), then their sum $\sum_{i=1}^{n} V_i = V_1+\cdots+V_n$ (which is a submodule of V) is also Noetherian (respectively, Artinian).*

In particular, if $V = V_1+\cdots+V_n$ and all $V_1,\dots,V_n$ are Noetherian (respectively, Artinian), then V itself is also Noetherian (respectively, Artinian).

Proof. Consider the natural surjective A-module homomorphism $g : V_1 \oplus\cdots\oplus V_n \to V_1+\cdots+V_n$. Let $W = \ker g$. This gives rise to the short exact sequence:

$$0 \to W \xrightarrow{i} V_1 \oplus\cdots\oplus V_n \xrightarrow{g} V_1+\cdots+V_n \to 0$$

By the third corollary, the finite direct sum $V_1 \oplus\cdots\oplus V_n$ is Noetherian. By the First Isomorphism Theorem, we have the isomorphism:

$$V_1+\cdots+V_n \cong (V_1 \oplus\cdots\oplus V_n)/W$$

Applying our initial lemma to the Noetherian module $V_1 \oplus\cdots\oplus V_n$ and its submodule W, we conclude that the quotient module $(V_1 \oplus\cdots\oplus V_n)/W$ must be Noetherian. Therefore, the sum $V_1+\cdots+V_n$ is also Noetherian. The proof for the Artinian case follows the exact same logic. □

4.4 Hilbert's Basis Theorem

One of the most foundational theorems in commutative algebra ensures that polynomial rings inherit the Noetherian property from their base rings.

Theorem 4.4.1 (Hilbert's Basis Theorem)**.** *If A is a Noetherian ring, then the polynomial ring $A[x]$ is also Noetherian.*

Proof. It is enough to prove the case $n = 1$, let $B = A[x]$.

To prove: every ideal I in B is finitely generated.

Let $I \subseteq B$ be any ideal, assume $I \neq 0, I \neq B$ (if equal, then it is generated by 0 or 1). Suppose I is not finitely generated.

Choose $0 \neq f_1 \in I$, such that $\deg f_1$ is least. *(Note: By the well-ordering principle of $\mathbb{N}$, any non-empty subset of $\mathbb{N}$ has a least element. Here, the set is $\Phi = \{\deg f \mid f \in I\setminus\{0\}\} \subseteq \mathbb{N}$).*

If $I\setminus\langle f_1\rangle \neq \emptyset$, choose $0 \neq f_2 \in I\setminus\langle f_1\rangle$, such that $\deg f_2$ is least. *(Since we assumed I is not finitely generated, it cannot be generated by just f_1).*

$\forall k \geq 1$, we have $f_{k+1} \in I\setminus\langle f_1,\dots,f_k\rangle$ such that $\deg f_{k+1}$ is least.

We can write these polynomials as:

$$\begin{aligned} f_1 &= a_1 x^{d_1} + \text{lower degree terms} \\ f_2 &= a_2 x^{d_2} + \text{lower degree terms} \\ &\vdots \\ f_k &= a_k x^{d_k} + \text{lower degree terms} \\ f_{k+1} &= a_{k+1} x^{d_{k+1}} + \text{lower degree terms} \end{aligned}$$

Since A is Noetherian, we can form an ascending chain of ideals in A:

$$\langle a_1 \rangle \subseteq \langle a_1, a_2 \rangle \subseteq \cdots \subseteq \langle a_1, \dots, a_k \rangle = \langle a_1, \dots, a_{k+1} \rangle = \dots$$

Because the chain stabilizes, there exists $k \geq 1$ such that:

$$b_1 a_1 + \cdots + b_k a_k = a_{k+1} \quad \text{for some } b_1, \dots, b_k \in A$$

Let us define a polynomial g:

$$g = b_1 x^{d_{k+1} - d_1} f_1 + b_2 x^{d_{k+1} - d_2} f_2 + \cdots + b_k x^{d_{k+1} - d_k} f_k - f_{k+1}$$

The coefficient of $x^{d_{k+1}}$ in g is:

$$(b_1 a_1 + b_2 a_2 + \cdots + b_k a_k) - a_{k+1} = 0$$

Thus, $\deg g < d_{k+1}$ and $g \in I \setminus \langle f_1, \dots, f_k \rangle$. *(Note: if $g \in \langle f_1, \dots, f_k \rangle$, then $f_{k+1} \in \langle f_1, \dots, f_k \rangle$, which is a contradiction `).*

On the other hand, $\deg g < \deg f_{k+1}$, which is a contradiction (`) to our choice of f_{k+1} having the least degree.

$\implies I$ must be finitely generated. $\square$

4.5 Consequences of the Basis Theorem

Corollary 4.5.1. *Let A be a Noetherian ring. Then every A-algebra of finite type over A is also a Noetherian ring.*

Proof. If B is an A-algebra of finite type, there exists a surjective A-algebra homomorphism:

$$\phi : A[x_1, \dots, x_n] \twoheadrightarrow B$$

By the Correspondence Theorem, the set of ideals of B, denoted $\mathfrak{I}(B)$, is in bijection with the set of ideals of $A[x_1, \dots, x_n]$ that contain the kernel of ϕ:

$$\mathfrak{I}(B) \cong \{ \mathfrak{a} \in \mathfrak{I}(A[x_1, \dots, x_n]) \mid \mathfrak{a} \supseteq \ker \phi \}$$

Since A is Noetherian, $A[x_1, \dots, x_n]$ is Noetherian by Hilbert's Basis Theorem. Therefore, every ideal in $A[x_1, \ldots, x_n]$ is finitely generated. It follows that every ideal in B is finitely generated, and hence B is Noetherian. $\square$

Corollary 4.5.2. *Every K-algebra of finite type, where K is a field, is a Noetherian ring.*

Corollary 4.5.3. *Every finite type $\mathbb{Z}$-algebra (or more generally, an algebra of finite type over any Principal Ideal Domain) is a Noetherian ring.*

CHAPTER 5

LOCALIZATION, RADICALS, AND THE KRULL-NAKAYAMA LEMMA

5.1 Rings and Modules of Fractions

Localization is an algebraic method of introducing denominators to a ring, formally inverting a chosen subset of elements. Let S be a multiplicatively closed subset of a ring A containing 1. We define an equivalence relation on $A \times S$ by $(a,s) \sim (b,t)$ if there exists $u \in S$ such that $u(at - bs) = 0$. The set of equivalence classes forms a ring $S^{-1}A$, the ring of fractions.

If $S = A \setminus \mathfrak{p}$ where $\mathfrak{p}$ is a prime ideal, $S^{-1}A$ is denoted $A_{\mathfrak{p}}$. This ring has a unique maximal ideal, making it a local ring.

Theorem 5.1.1 (Universal Property of S^{-1} (More General))**.** *Let $\phi : A \to B$ be a ring homomorphism, and $S \subseteq A$ a multiplicatively closed subset such that $\phi(S) \subseteq B^{\times}$. Further, let V be an A-module and W be a B-module, and let $f : V \to W$ be an A-module homomorphism.*

(Note: W is considered an A-module by restriction of scalars via ϕ. W is also considered an $S^{-1}A$-module by restriction of scalars using the ring homomorphism $S^{-1}A \to B$).

Then there exists a unique $S^{-1}A$-module homomorphism $S^{-1}f : S^{-1}V \to W$ such that $f = (S^{-1}f) \circ i_S^V$. This means the following diagram is commutative:

$$\begin{tikzcd}
V \arrow[rr, "f"] \arrow[dr, "i_S^V"'] & & W \\
& S^{-1}V \arrow[ur, dashed, "S^{-1}f"'] &
\end{tikzcd}$$

Furthermore, the kernel is given by:

$$\ker(S^{-1}f) = \{x/s \mid x \in \ker(f), s \in S\}$$

Similarly, for an A-module V, one can define the module of fractions $S^{-1}V$. This construction satisfies a fundamental universal property regarding module homomorphisms.

Theorem 5.1.2 (Universal Property of S^{-1} on Modules)**.** *Let A be a ring, $S \subseteq A$ a multiplicatively closed subset, let V,W be two A-modules, and let $f : V \to W$ be an A-module homomorphism. There exists a unique $S^{-1}A$-module homomorphism $S^{-1}f : S^{-1}V \to S^{-1}W$ such that the following diagram is commutative:*

$$\begin{tikzcd} V \arrow[r, "f"] \arrow[d, "i_S^V"'] & W \arrow[d, "i_S^W"] \\ S^{-1}V \arrow[r, dashed, "S^{-1}f"'] & S^{-1}W \end{tikzcd}$$

which implies $i_S^W \circ f = S^{-1}f \circ i_S^V$.

Specifically, the map $S^{-1}f$ *demands that for all* $x \in V$*:*

$$x \longmapsto f(x) \longmapsto f(x)/1$$
$$x \longmapsto x/1 \longmapsto S^{-1}f(x/1)$$

5.2 The Local Global Principle

Many properties of a module can be checked "locally" at prime or maximal ideals. This algebraic principle heavily mimics the geometric idea of checking properties of a space locally in the neighborhoods of points.

Theorem 5.2.1 (Local Global Principle)**.** *Let A be a ring, and V be an A-module. The following are equivalent (TFAE):*

1. $V = 0$
2. $V_{\mathfrak{p}} = 0$ *(where* $V_{\mathfrak{p}} := (A \setminus \mathfrak{p})^{-1}V$*) for every* $\mathfrak{p} \in Spec(A)$
3. $V_{\mathfrak{m}} = 0 \quad \forall \mathfrak{m} \in Spm(A)$

5.3 Radicals of Ideals and Rings

Definition 5.3.1 (Radicals and Nilpotent Ideals)**.** Let A be a commutative ring.

- An ideal $\mathfrak{a}$ in A is called **nilpotent** if $\mathfrak{a}^r = 0$ for some integer $r \geq 1$.
- The **radical** of an ideal I, denoted $\sqrt{I}$, is defined as $\{a \in A \mid a^n \in I \text{ for some integer } n \geq 1\}$.
- The set of all nilpotent elements in A is denoted $\mathrm{nil}(A)$. If $\mathrm{nil}(A)$ is finitely generated, then it is a nilpotent ideal.
- The **nilradical** of A, denoted $\mathfrak{n}_A$, is the intersection of all prime ideals in A:

$$\mathfrak{n}_A = \bigcap_{\mathfrak{p} \in \mathrm{Spec}(A)} \mathfrak{p}$$

  Note that $\mathfrak{n}_A = \mathrm{nil}(A) = \sqrt{\langle 0 \rangle}$.

- The **Jacobson radical** of A, denoted $\mathfrak{m}_A$ (or $J(A)$), is the intersection of all maximal ideals in A:

$$\mathfrak{m}_A = \bigcap_{\mathfrak{m} \in \mathrm{Spm}(A)} \mathfrak{m}$$

It follows directly from the definitions that $\mathfrak{n}_A = \sqrt{\langle 0 \rangle} \subseteq \mathfrak{m}_A$. If $\mathfrak{n}_A = 0$, the ring A is called a **reduced** ring.

Example 5.3.2. We observe the relationship between the nilradical and Jacobson radical in various rings:

1. In the ring of integers $\mathbb{Z}$, the nilradical $\mathfrak{n}_{\mathbb{Z}} = 0$. The Jacobson radical $\mathfrak{m}_{\mathbb{Z}} = \bigcap_{p \in \mathscr{P}} \langle p \rangle = 0$. In general, $\mathfrak{n}_A = 0$ for any integral domain.

2. Let K be a field and $R = K[X_1, \ldots, X_n]$. Since R is an integral domain, $\mathfrak{n}_R = 0$. Furthermore, $\mathfrak{m}_R = 0$.

3. Let K be a field and consider the formal power series ring $K[[X]]$. This is a local ring with a unique maximal ideal generated by X. Thus, $\mathfrak{m}_{K[[X]]} = \langle X \rangle$. However, being an integral domain, its nilradical is $\mathfrak{n}_{K[[X]]} = 0$. This provides an example where $\mathfrak{n}_A \neq \mathfrak{m}_A$.

Theorem 5.3.3. *Let A be an Artinian commutative ring. Then the following properties hold:*

1. *The Jacobson radical $\mathfrak{m}_A$ of A is nilpotent, meaning $\mathfrak{m}_A^n = 0$ for some integer $n \geq 1$.*

2. *The maximal spectrum $Spm(A)$ is a finite set.*

3. *A is a Noetherian ring.*

Corollary 5.3.4. *Let K be a field and A be a finite K-algebra such that its vector space dimension is finite ($\dim_K A = n < \infty$). Then the number of maximal ideals and K-rational points are bounded by this dimension:*

$$\#Spm(A) \leq \dim_K A \quad \textit{and} \quad \#K\text{-}Spec(A) \leq \dim_K A$$

5.4 The Krull-Nakayama Lemma

The Krull-Nakayama Lemma is a highly versatile tool in commutative algebra, particularly when working with finitely generated modules over local rings.

Lemma 5.4.1 (Krull-Nakayama Lemma)**.** *Let A be a ring and $\mathfrak{a} \in \mathfrak{I}(A)$ be an ideal. The following are equivalent (TFAE):*

1. *$\mathfrak{a} \subseteq \mathfrak{m}_A$ (the ideal is contained in the Jacobson radical of A).*

2. *For every finite A-module V, if $V = \mathfrak{a}V$, then $V = 0$.*

Corollary 5.4.2. *Let $U \subseteq V$ be A-modules such that the quotient module V/U is a finite A-module. Let $\mathfrak{a} \subseteq \mathfrak{m}_A$ be an ideal. If $U + \mathfrak{a}V = V$, then $V = U$.*

Corollary 5.4.3. *Let $f : V \to W$ be an A-module homomorphism such that its cokernel, coker$(f) = W/Img(f) = W/f(V)$, is a finite A-module. Further, let $\mathfrak{a} \subseteq \mathfrak{m}_A$ be an ideal such that the induced $A/\mathfrak{a}$-module homomorphism:*

$$\overline{f} : V/\mathfrak{a}V \to W/\mathfrak{a}W$$

defined by $\bar{x} \longmapsto \overline{f(x)}$ is surjective. Then the original homomorphism f is also surjective.

CHAPTER 6

HILBERT'S NULLSTELLENSATZ

6.1 Hilbert's Nullstellensatz (Zero Place Theorem)

Before stating the main theorem, we establish two important algebraic lemmas.

Lemma 6.1.1. *Suppose that $A \subseteq R \subseteq B$ is a sequence of ring extensions, where A is Noetherian and B is an A-algebra of finite type. If B is a finite R-module (meaning finitely many elements can be written as a linear combination of generators), then R is an A-algebra of finite type. In particular, R is a Noetherian ring by Hilbert's Basis Theorem.*

Lemma 6.1.2. *Let $Y_1, \ldots, Y_m$ (for $m \geq 1$) be indeterminates over a field K. Then the field of rational functions $K(Y_1, \ldots, Y_m) = Qt(K[Y_1, \ldots, Y_m])$ is NOT a K-algebra of finite type.*

Let L/K be a field extension where L is algebraically closed. Let $R = K[X_1, \ldots, X_n]$ be the polynomial ring, and let $\mathfrak{a}$ be an ideal in R, generated by finitely many polynomials such that $\mathfrak{a} = \langle f_1, \ldots, f_m \rangle$. The vanishing locus $V_L(\mathfrak{a}) \subseteq L^n$ forms a K-algebraic subset.

The Nullstellensatz can be stated in three equivalent formulations:

Theorem 6.1.3 (HNS 1: Classical Formulation)**.** *If L is algebraically closed and $\mathfrak{a} \neq R$, then $V_L(\mathfrak{a}) \neq \emptyset$.*

Theorem 6.1.4 (HNS 2: Geometric Formulation)**.** *If L is algebraically closed, then for every ideal $\mathfrak{a}$ in R, the ideal of the vanishing locus is equal to the radical of the ideal:*

$$I_K(V_L(\mathfrak{a})) = \sqrt{\mathfrak{a}}$$

Theorem 6.1.5 (HNS 3: Algebraic Formulation / Zariski's Lemma)**.** *Let K/k be a field extension. If K is a k-algebra of finite type, meaning $K = k[x_1, \ldots, x_n] \cong k[X_1, \ldots, X_n]/\mathfrak{a}$, then the extension K/k is algebraic.*

Consequences of Hilbert's Nullstellensatz

Let L/K be a field extension and $R = K[X_1, \ldots, X_n]$. The Nullstellensatz yields several profound corollaries that cement the algebraic-geometric dictionary.

Corollary 6.1.6. *If L is algebraically closed and $\mathfrak{a}_1, \mathfrak{a}_2 \in \mathfrak{I}(R)$, then:*

$$V_L(\mathfrak{a}_1) = V_L(\mathfrak{a}_2) \iff \sqrt{\mathfrak{a}_1} = \sqrt{\mathfrak{a}_2}$$

Corollary 6.1.7. *If L is algebraically closed, then the map from the set of K-algebraic sets in L^n to r-$\mathfrak{I}(R)$ (the set of radical ideals in R) given by $V \mapsto I_K(V)$ is bijective. Furthermore, we have the equalities:*

$$I_K(V_L(\mathfrak{a})) = \sqrt{\mathfrak{a}} \quad \text{and} \quad I_K\left(\bigcap_{i\in I} V_i\right) = \sqrt{\sum_{i\in I} I_K(V_i)}$$

For a point $a = (a_1, \dots, a_n) \in L^n$, the ideal $I_K(\{a\}) = \{f \in R \mid f(a) = 0\}$, denoted simply as $I_K(a)$, is a radical ideal in R.

Definition 6.1.8. For a field extension L/K and a point $a = (a_1, \dots, a_n) \in L^n$, we say that a is **algebraic over** K if each coordinate a_i (for $i = 1, \dots, n$) is algebraic over K.

Corollary 6.1.9. *With the above notations, for a point $a \in L^n$, the ideal $I_K(a)$ is a maximal ideal in R if and only if a is algebraic over K.*

Definition 6.1.10. Two K-algebraic points $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n)$ in L^n are called K**-conjugates** (or conjugates over K) if there exists a K-algebra isomorphism $\alpha : K[a_1, \dots, a_n] \xrightarrow{\approx} K[b_1, \dots, b_n]$ such that $\alpha(a_i) = b_i$ for all $i = 1, \dots, n$. Note that if $a, b \in K^n$, then a and b are K-conjugate if and only if $a_i = b_i$ for all $i = 1, \dots, n$.

Corollary 6.1.11. *Let $S = \{a = (a_1, \dots, a_n) \in L^n \mid a_1, \dots, a_n$ are algebraic over $K\}$. The map $\phi : S \to Spm(R)$ defined by $a \mapsto I_K(a)$ is surjective if L is algebraically closed. Moreover, for two K-algebraic points $a, b \in L^n$, we have $\phi(a) = \phi(b)$ if and only if a is a K-conjugate to b.*

Corollary 6.1.12 (HNS 4: Classical)**.** *If $K = L$ is algebraically closed, then the mapping $\phi : K^n \to Spm(K[X_1, \dots, X_n])$ defined by $a = (a_1, \dots, a_n) \mapsto \mathfrak{m}_a = \langle X_1 - a_1, \dots, X_n - a_n \rangle$ is bijective. Moreover, for any ideal $\mathfrak{a} \subseteq K[X_1, \dots, X_n]$, we have:*

$$\phi(V_K(\mathfrak{a})) = \{\mathfrak{m} \in Spm(K[X_1, \dots, X_n]) \mid \mathfrak{a} \subseteq \mathfrak{m}\}$$

Corollary 6.1.13. *Let K be a field, $\mathfrak{a} \in \mathfrak{I}(K[X_1, \dots, X_n])$, and E/K be any field extension. Then $\mathfrak{a}E[X_1, \dots, X_n] \neq E[X_1, \dots, X_n]$ if $\mathfrak{a} \neq K[X_1, \dots, X_n]$.*

Corollary 6.1.14. *Let K be a field and A be a K-algebra of finite type. For any ideal $\mathfrak{a} \in \mathfrak{I}(A)$:*

$$\sqrt{\mathfrak{a}} = \bigcap_{\substack{\mathfrak{m}\in Spm(A) \\ \mathfrak{a}\subseteq\mathfrak{m}}} \mathfrak{m}$$

In particular, the nilradical $\mathfrak{n}_A$ equals the Jacobson radical $\mathfrak{m}_A$:

$$\mathfrak{n}_A = \sqrt{\langle 0 \rangle} = \bigcap_{\mathfrak{m}\in Spm(A)} \mathfrak{m} = \mathfrak{m}_A$$

CHAPTER 7

THE ALGEBRA-GEOMETRY DICTIONARY AND ZARISKI TOPOLOGY

7.1 Properties of the Vanishing Locus and Ideal

Let $R = K[X_1, \ldots, X_n]$ be the polynomial ring over a field K, and let L/K be a field extension. We define the vanishing ideal map I_K from the power set of L^n to the set of radical ideals in R, $r\text{-}\mathfrak{I}(R)$:

$$I_K : \mathscr{P}(L^n) \longrightarrow r\text{-}\mathfrak{I}(R)$$

$$Y \longmapsto I_K(Y) := \{f \in R \mid f(y) = 0 \text{ for all } y \in Y\}$$

Note that $I_K(Y) = \bigcap_{y \in Y} I_K(y)$, making it a radical ideal in R.

The mappings V_L and I_K satisfy the following fundamental properties:

Proposition 7.1.1 (Algebra-Geometry Dictionary)**.** *Let $\mathfrak{a}, \mathfrak{b} \in \mathfrak{I}(R)$ and let $Y, Y' \subseteq L^n$.*

1. ***Trivial Sets:*** *$V_L(\emptyset) = V_L(0) = L^n$ and $V_L(R) = V_L(1) = \emptyset$. (These establish the empty set and full space for the Zariski topology).*

2. ***Ideals of the Full Space:*** *$I_K(\emptyset) = R$. Furthermore, $I_K(L^n) = 0$ if L is an infinite field. However, if K is a finite field with $\#K = q$, then $I_K(K^n) = \langle X_1^q - X_1, \ldots, X_n^q - X_n \rangle$.*

3. ***Inclusion Reversing:*** *Both V_L and I_K reverse inclusions:*

$$\mathfrak{a} \subseteq \mathfrak{b} \implies V_L(\mathfrak{a}) \supseteq V_L(\mathfrak{b})$$
$$Y \subseteq Y' \subseteq L^n \implies I_K(Y) \supseteq I_K(Y')$$

4. ***Closure Operators:*** *$(I_K \circ V_L)(\mathfrak{a}) \supseteq \sqrt{\mathfrak{a}}$. Furthermore, $(V_L \circ I_K)(Y) = Y$ if and only if Y is a K-algebraic subset in L^n.*

5. ***Equality of Sets:*** *If V_1 and V_2 are two K-algebraic sets, then $V_1 = V_2 \iff I_K(V_1) = I_K(V_2)$.*

6. ***Ideals of Unions and Intersections:*** *For a family of K-algebraic sets V_i ($i \in I$):*

$$I_K\left(\bigcup_{i \in I} V_i\right) = \bigcap_{i \in I} I_K(V_i) \quad \text{and} \quad I_K\left(\bigcap_{i \in I} V_i\right) \supseteq \sqrt{\sum_{i \in I} I_K(V_i)}$$

7. ***Loci of Sums and Products:*** *For ideals* $\mathfrak{a}_i \in \mathfrak{I}(R)$*:*

$$V_L\left(\bigcap_{i=1}^{m} \mathfrak{a}_i\right) = V_L(\mathfrak{a}_1 \cdots \mathfrak{a}_m) = \bigcup_{i=1}^{m} V_L(\mathfrak{a}_i)$$

$$V_L\left(\sum_{i\in I} \mathfrak{a}_i\right) = \bigcap_{i\in I} V_L(\mathfrak{a}_i)$$

(Note: Properties 1 and 7 explicitly verify the axioms required to form the Zariski Topology).

8. ***Rational Points:*** *For every point* $a = (a_1, \dots, a_n) \in K^n \subseteq L^n$*, the vanishing ideal is a maximal ideal in R:*

$$I_K(\{a\}) = \langle X_1 - a_1, \dots, X_n - a_n \rangle = \mathfrak{m}_a \in Spm(R)$$

7.2 The Zariski Topology

7.2.1 The Geometric Zariski K-Topology

We consider L^n with the Zariski Topology, or equivalently $\mathbb{A}_L^n$ (affine n-space over L). The Zariski K-Topology on L^n is defined by taking the closed sets to be the affine K-algebraic sets. Let the family of closed sets be:

$$\mathscr{F} := \{V_L(I) \mid I \subseteq K[X_1, \dots, X_n] \text{ is an ideal (or radical ideal)}\}$$

We must verify that this forms a valid topology by checking the three topological axioms for closed sets:

1. **Empty set and full space:** The empty set and the entire space are closed.

$$\emptyset = V_L(1) = V_L(K[X_1, \dots, X_n]) = V_L(\langle 1 \rangle) = V_L(\langle a \rangle) \text{ for } a \in K \setminus \{0\}$$

$$L^n = V_L(0)$$

2. **Arbitrary Intersections:** Let $V_L(I_i)$ be a collection of closed sets for $i \in B$, where $I_i \subseteq K[X_1, \dots, X_n]$.

$$\begin{aligned} a \in \bigcap_{i\in B} V_L(I_i) &\iff a \in V_L(I_i) \text{ for all } i \\ &\iff f(a) = 0 \text{ for all } f \in I_i, \text{ for all } i \\ &\iff f(a) = 0 \text{ for all } f \in \sum_{i\in B} I_i \\ &\iff a \in V_L\left(\sum_{i\in B} I_i\right) \end{aligned}$$

Thus, arbitrary intersections of closed sets are closed.

3. **Finite Unions:** We must show $V_L(I_1) \cup V_L(I_2) = V_L(I_1 \cap I_2) = V_L(I_1 I_2)$.

First, observe that the V_L operation reverses inclusions:

$$I_1 \cap I_2 \subseteq I_1 \implies V_L(I_1) \subseteq V_L(I_1 \cap I_2)$$
$$I_1 \cap I_2 \subseteq I_2 \implies V_L(I_2) \subseteq V_L(I_1 \cap I_2)$$

This implies $V_L(I_1) \cup V_L(I_2) \subseteq V_L(I_1 \cap I_2)$. Furthermore, since $I_1 I_2 \subseteq I_1 \cap I_2$, we have $V_L(I_1 \cap I_2) \subseteq V_L(I_1 I_2)$.

Now, to show the reverse direction: If $a \in V_L(I_1)$, then $f(a) = 0$ for all $f \in I_1$. For any $F \in I_1, G \in I_2$, we have $(FG)(a) = F(a)G(a) = 0 \cdot G(a) = 0$. So $a \in V_L(I_1 I_2)$. The same holds for $a \in V_L(I_2)$. This gives $V_L(I_1) \cup V_L(I_2) \subseteq V_L(I_1 I_2)$.

Finally, let $a \in V_L(I_1 I_2)$. Suppose for contradiction that $a \notin V_L(I_1)$ and $a \notin V_L(I_2)$. Then:

$$\exists F \in I_1 \text{ such that } F(a) \neq 0$$
$$\exists G \in I_2 \text{ such that } G(a) \neq 0$$

This implies $H(a) = F(a)G(a) \neq 0$, which contradicts $a \in V_L(I_1 I_2)$. Therefore, $V_L(I_1 I_2) \subseteq V_L(I_1) \cup V_L(I_2)$.

Conclusively, the mapping defines a bijective correspondence between radical ideals and closed geometric sets:

$$V_L : r\text{-}\mathfrak{I}(K[X_1, \ldots, X_n]) \longrightarrow \{\text{closed subsets in } \mathbb{A}^n_L \text{ wrt Zariski Topology}\}$$

7.2.2 *Zariski Topology on the Spectrum of a Ring*

We can generalize this topology from affine spaces to arbitrary commutative rings. Let A be a commutative ring and $X = \operatorname{Spec}(A)$. The collection

$$\tau_X = \{V(\mathfrak{a}) \mid \mathfrak{a} \in \mathfrak{I}(A)\}$$

satisfies the properties of closed subsets in a topological space. In particular, the pair (X, τ_X) is a topological space with τ_X as its closed subsets. This is called the **Zariski Topology** on $X = \operatorname{Spec}(A)$.

The open subsets in X are precisely the complements of the closed sets. We denote these as:

$$D(\mathfrak{a}) := X \setminus V(\mathfrak{a}) \quad \text{for } \mathfrak{a} \in \mathfrak{I}(A)$$

For a single element $f \in A$, we define the principal (or distinguished) open sets:

$$D(f) = D(\langle f \rangle) = X \setminus V(f) = \{\mathfrak{p} \in X \mid f \notin \mathfrak{p}\}$$

If A is a Noetherian ring, any open set can be expressed as a finite union of these principal open sets:

$$D(f_1, \ldots, f_m) = X \setminus V(f_1, \ldots, f_m) = X \setminus \bigcap_{j=1}^{m} V(f_j) = \bigcup_{j=1}^{m} D(f_j)$$

CHAPTER 8

INTEGRAL EXTENSIONS AND NOETHER NORMALIZATION LEMMA

8.1 Integral Extensions

Let A be a base commutative ring with identity 1_A. Let B be an A-algebra with the structure homomorphism $\phi : A \to B$. Note that B may be non-commutative, in which case we require $\phi(A) \subseteq Z(B)$, the center of B.

Recall that given an element $x \in B$, there exists a unique A-algebra homomorphism, called the substitution or evaluation map:

$$\varepsilon_x : A[X] \to A[x] \subseteq B$$

where $X \mapsto x$ and $F(X) \mapsto F(x)$. Here, $A[x]$ is the A-subalgebra of B generated by x. The kernel of this map, $\ker \varepsilon_x$, is an ideal in $A[X]$. By the First Isomorphism Theorem, we have:

$$A[X]/\ker \varepsilon_x \cong A[x] \subseteq B$$

Depending on the nature of $\ker \varepsilon_x$, we classify the element x into one of three categories:

1. If $\ker \varepsilon_x = \{0\}$, then $A[X] \cong A[x]$. We say x is **algebraically independent** or **transcendental** over A.

2. If $\ker \varepsilon_x \neq \{0\}$, there exists some non-zero polynomial $F \in \ker \varepsilon_x$ such that $F(x) = 0$. In this case, we say x is **algebraic** over A.

3. If $\ker \varepsilon_x$ contains a **monic** polynomial $f \in A[X]$, we say x is **integral** over A.

More explicitly, let R and S be commutative rings where $R \subseteq S$. An element $s \in S$ is integral over R if it is the root of a monic polynomial with coefficients strictly belonging to R. This means there exist elements $r_0, r_1, \dots, r_{n-1} \in R$ such that:

$$s^n + r_{n-1}s^{n-1} + \cdots + r_1 s + r_0 = 0$$

The crucial condition here is that the polynomial is monic—the leading coefficient of s^n must be exactly 1.

Definition 8.1.1 (Integral Extension)**.** If every single element of the ring S is integral over R, then we say that S is an **integral extension** of R.

Note: If the base ring $A = K$ is a field, then all ideals in $K[X]$ are principal. Because of this, the concepts of being "algebraic over K" and "integral over K" are mathematically equivalent. Integral extensions preserve dimension and play a crucial role in relating the geometry of affine varieties under finite morphisms.

8.2 Noether Normalization Lemma

A pivotal result bridging integral extensions and the geometric structure of affine varieties is the Noether Normalization Lemma.

Lemma 8.2.1 (Prerequisite Lemma)**.** *Let K be a field, and $F \in K[X_1, X_2, \ldots, X_n]$ be a non-constant polynomial. Then there exists a K-algebra automorphism:*

$$\phi : K[X_1, \ldots, X_n] \xrightarrow[\textit{K-alg Auto}]{\approx} K[X_1, \ldots, X_n]$$

such that $\phi(X_n) = X_n$ and

$$F = aX_n^d + f_{d-1}X_n^{d-1} + \cdots + f_1 X_n + f_0$$

where $a \in K^\times$, and $f_j \in K[Y_1, \ldots, Y_{n-1}]$ for $j = 0, \ldots, d-1$, with $Y_i = \phi(X_i)$ for $i = 1, \ldots, n-1$.

Theorem 8.2.2 (Noether Normalization Lemma)**.** *Let A be a K-algebra of finite type, meaning there exists a surjective homomorphism:*

$$K[X_1, \ldots, X_n] \xrightarrow[\textit{surj}]{\textit{K-alg homo}} A = K[x_1, \ldots, x_n]$$

Then there exist $z_1, z_2, \ldots, z_m \in A$, such that:

1. *$z_1, \ldots, z_m$ are algebraically independent over K. i.e.,*

$$K[Z_1, Z_2, \ldots, Z_m] \xrightarrow[\approx]{\varepsilon_\alpha} K[z_1, \ldots, z_m] \subseteq A$$

   via the mapping $Z_i \longmapsto z_i$.

2. *A is integral over $K[z_1, \ldots, z_m]$.*

Furthermore, if $x_1, \ldots, x_n$ are algebraically dependent over K, then $m < n$.

CHAPTER 9

CONCLUSION

This summer internship served as an intensive exploration into the intricate structural relationship between Commutative Algebra and Algebraic Geometry. Beginning with a rigorous algebraic foundation—examining the elemental properties of rings, ideals, and localizing structures—the progression clearly illuminated how chain conditions uniquely govern the complexity of abstract mathematical spaces. The deep investigations into the properties of localization, radical ideals, and the Krull-Nakayama Lemma fortified the algebraic machinery necessary to transition seamlessly into pure geometry. This systematic study culminated in the mastery of Hilbert's Nullstellensatz and the algebraic-geometric dictionary it forms. By formally defining the Zariski topology on topological spaces like spectra and characterizing affine dimension via the Noether Normalization Lemma, the project succeeded in bridging the gap between algebraic ideals and geometric points. The sophisticated theoretical mechanisms established in this report are absolutely vital for future endeavors into modern algebraic geometry, arithmetic geometry, and scheme theory.

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